S&S QUIZ-7 SOLUTION

Sol(1):- A causal & stable LTI system S has frequency response,

$$H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega}$$

(a) Given, $H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega}$

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{4+j\omega}{6-\omega^2+5j\omega}$$

$$6\cdot Y(j\omega) - \omega^2 \cdot Y(j\omega) + 5j\omega \cdot Y(j\omega) = 4X(j\omega) + j\omega \cdot X(j\omega)$$

$$6\cdot Y(j\omega) + (j\omega)^2 \cdot Y(j\omega) + 5j\omega \cdot Y(j\omega) = 4X(j\omega) + j\omega \cdot X(j\omega) \quad \text{---(1)}$$

As we know that-

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\cdot e^{j\omega t}\cdot d\omega \quad \text{---(2)}$$

where, $x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} X(j\omega)$

Differentiating both side w.r.t. 't', we get-

$$\frac{d}{dt}x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\cdot(j\omega)\cdot e^{j\omega t}\cdot d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} (j\omega)\cdot X(j\omega)\, e^{j\omega t}\cdot d\omega$$

which shows, $\frac{d}{dt}x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} (j\omega)\,X(j\omega)$ ---(3)

Similarly, $\frac{d^2}{dt^2}x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} (j\omega)^2 \cdot X(j\omega)$ ---(4)

→ (1 Mark)

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Taking inverse fourier transform (IFT) of both side of eqn(1) with the help of eqn(3) & eqn(4), we get –

$$6\,y(t) + \frac{d^2}{dt^2}y(t) + 5\frac{d}{dt}y(t) = 1\,x(t) + \frac{d}{dt}x(t)$$

$$\frac{d^2}{dt^2}y(t) + 5\frac{d}{dt}y(t) + 6\,y(t) = \frac{d}{dt}x(t) + 1\cdot x(t)$$

→ (2 Marks)

(b) Given – $H(j\omega) = \frac{4+j\omega}{6-\omega^2+5j\omega} = \frac{4+j\omega}{(2+j\omega)(3+j\omega)}$

$$= \frac{2}{2+j\omega} - \frac{1}{3+j\omega} \quad \text{—— (5)}$$

As we know that –

$$e^{-at}\,u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{1}{a+j\omega}, \quad a>0 \quad \text{—— (6)}$$

Taking inverse fourier transform (IFT) of both side of eqn (5) with the help of eqn(6), we get –

$$h(t) = 2e^{-2t}\cdot u(t) - e^{-3t}\cdot u(t)$$

→ (2 Marks)

(c) Given, $x(t) = e^{-4t}u(t) - t^2e^{-4t}u(t)$ —— (9)

As we know that –

$$t\cdot e^{-at}u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{1}{(a+j\omega)^2} \quad \text{—— (7)}$$

$$t^2\cdot e^{-at}\cdot u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{2}{(a+j\omega)^3} \quad \text{—— (8)}$$

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Taking FT of both side of eqn (9) with the help of eqn (6) & eqn (8), we get —

$$X(j\omega) = \frac{1}{4+j\omega} - \frac{2}{(4+j\omega)^3} = \frac{(4+j\omega)^2 - 2}{(4+j\omega)^3}$$

→ (1 Marks)

Therefore, $Y(j\omega) = H(j\omega) \cdot X(j\omega)$

$$= \frac{(4+j\omega)}{(2+j\omega)(3+j\omega)} \times \frac{(4+j\omega)^2 - 2}{(4+j\omega)^3}$$

$$= \frac{(4+j\omega)^2 - 2}{(2+j\omega)(3+j\omega)(4+j\omega)^2}$$

$$= \frac{1/2}{2+j\omega} + \frac{1}{3+j\omega} - \frac{3/2}{4+j\omega} - \frac{1}{(4+j\omega)^2}$$

—— (10)

Taking IFT of both side of eqn (10) with the help of eqn (6) & eqn (7), we get —

$$y(t) = \frac{1}{2} e^{-2t} u(t) + e^{-3t} u(t) - \frac{3}{2} e^{-4t} \cdot u(t) - t e^{-4t} u(t)$$

→ (2 Marks)

Proofs:

(I) $x(t) = e^{-at} u(t)$, $a > 0$

$$\therefore X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt$$

$$= \int_{0}^{\infty} e^{-(a+j\omega)t} dt = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{(a+j\omega)} \left[e^{-(a+j\omega)\infty} - e^{0} \right] = \frac{-1}{(a+j\omega)} (0-1)$$

$$= \left(\frac{1}{a+j\omega} \right)$$

→ (1 Marks)

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(II) $x(t) = t\, e^{-at} u(t)$, $a > 0$

As we know that — $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot (-jt)\, dt$$

$$j \frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} [t\, x(t)]\, e^{-j\omega t}\, dt$$

which shows —

$$t\, x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} j \frac{d}{d\omega} X(j\omega)$$

similarly, $t^n x(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} (j)^n \frac{d^n}{d\omega^n} X(j\omega)$

Hence, $t e^{-at} u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{1}{(a+j\omega)^2}$

$t^2 e^{-at} u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{2}{(a+j\omega)^3}$

$\vdots$

$t^n e^{-at} u(t) \underset{IFT}{\overset{FT}{\rightleftharpoons}} \frac{\lfloor n}{(a+j\omega)^{n+1}}$ → (1 Marks)

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